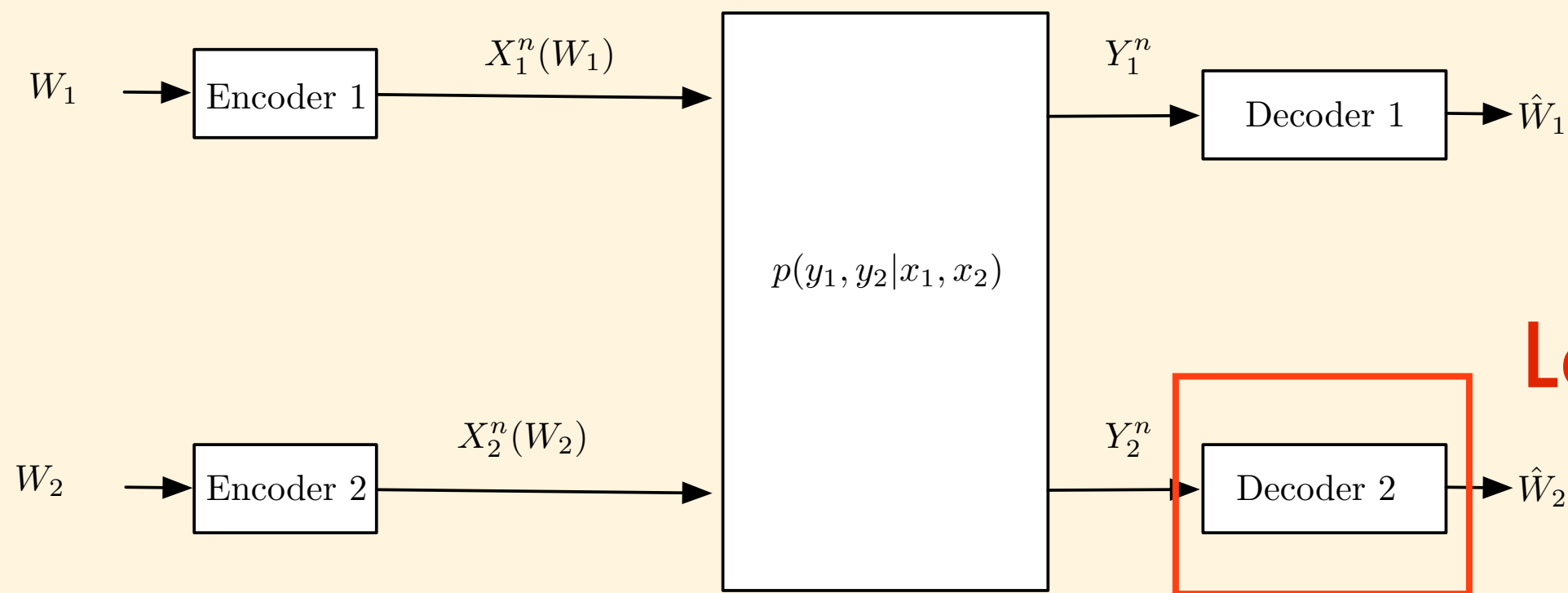


The Gaussian interference
channel with lack of codebook
knowledge at one receiver:
symmetric capacity to within a
gap with a PAM input

Alex Dytso, Daniela Tuninetti, Natasha Devroye

Motivation



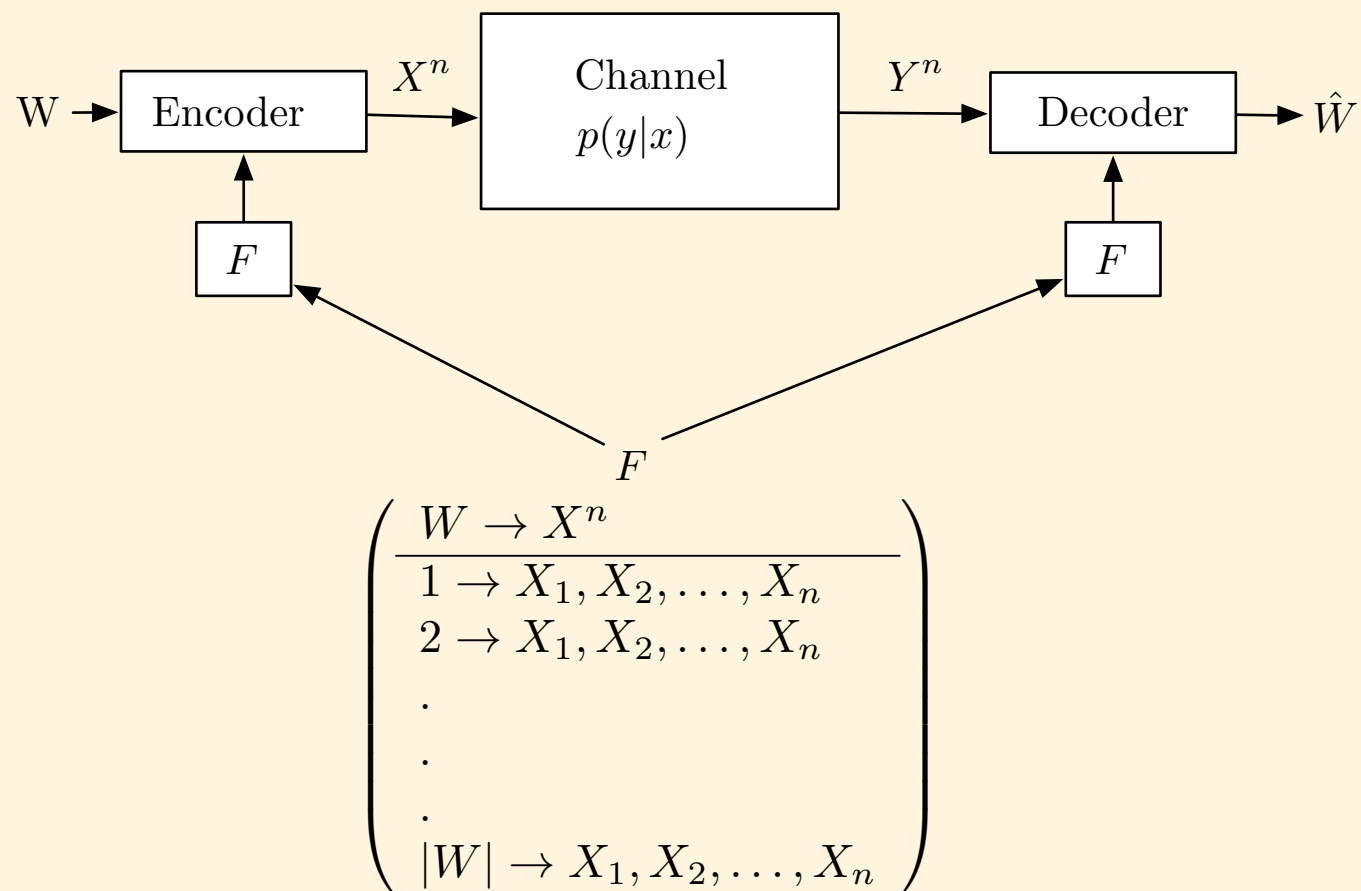
**Legacy Receiver
No Joint-
Decoding**

Outline

- Relevant Channel Models
- Past Work
- Generalized Ozarow-Wyner Bound
- Discrete inputs are ‘good’ inputs and ‘good’ interferers
- Capacity and Approximate Capacity Results

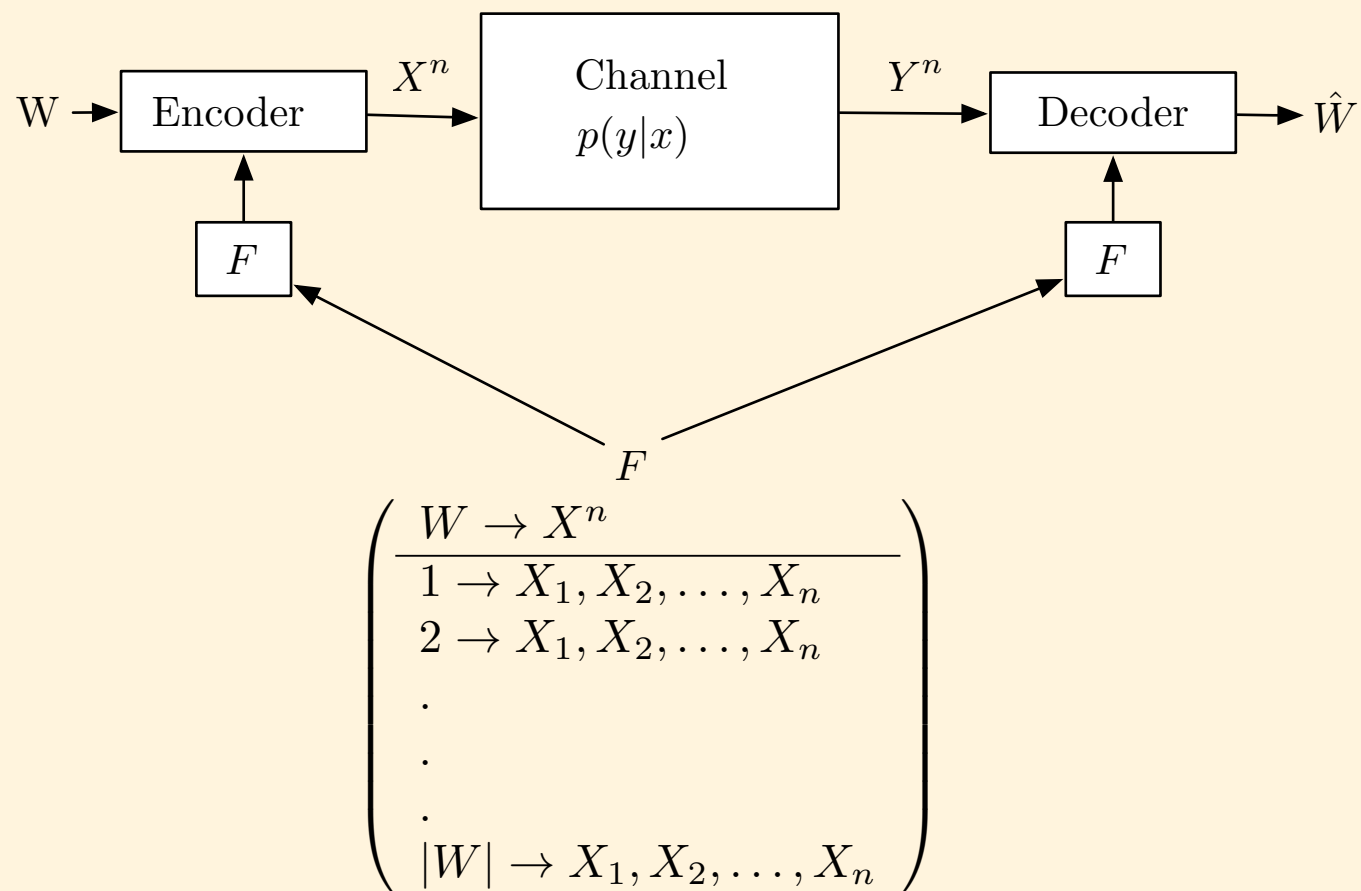
Codebook Knowledge

Point-to-Point Channel

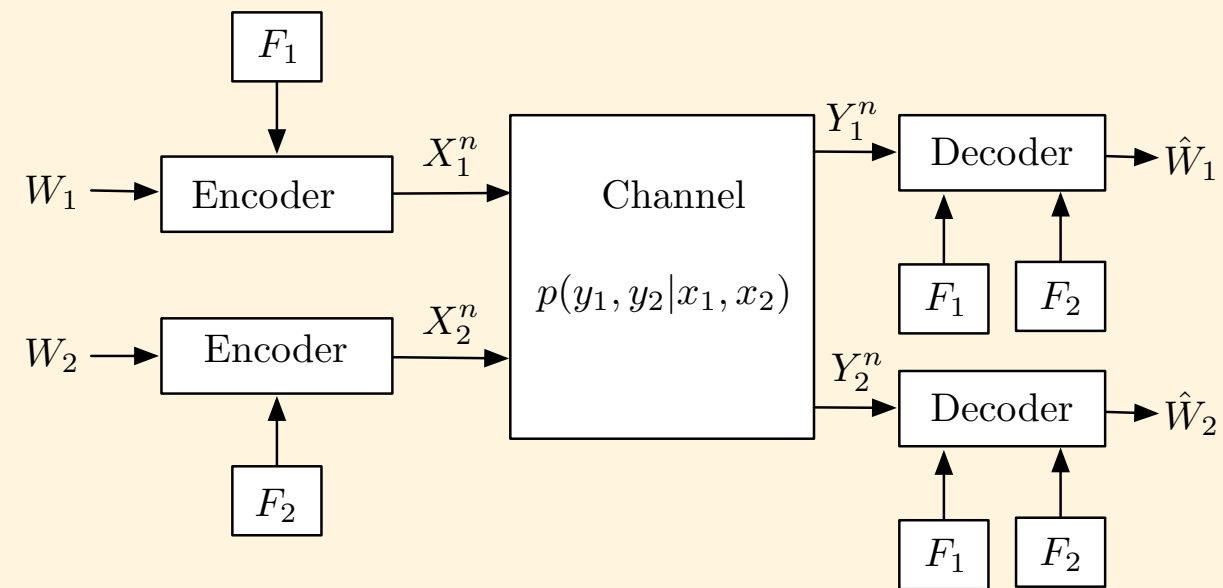


Codebook Knowledge

Point-to-Point Channel

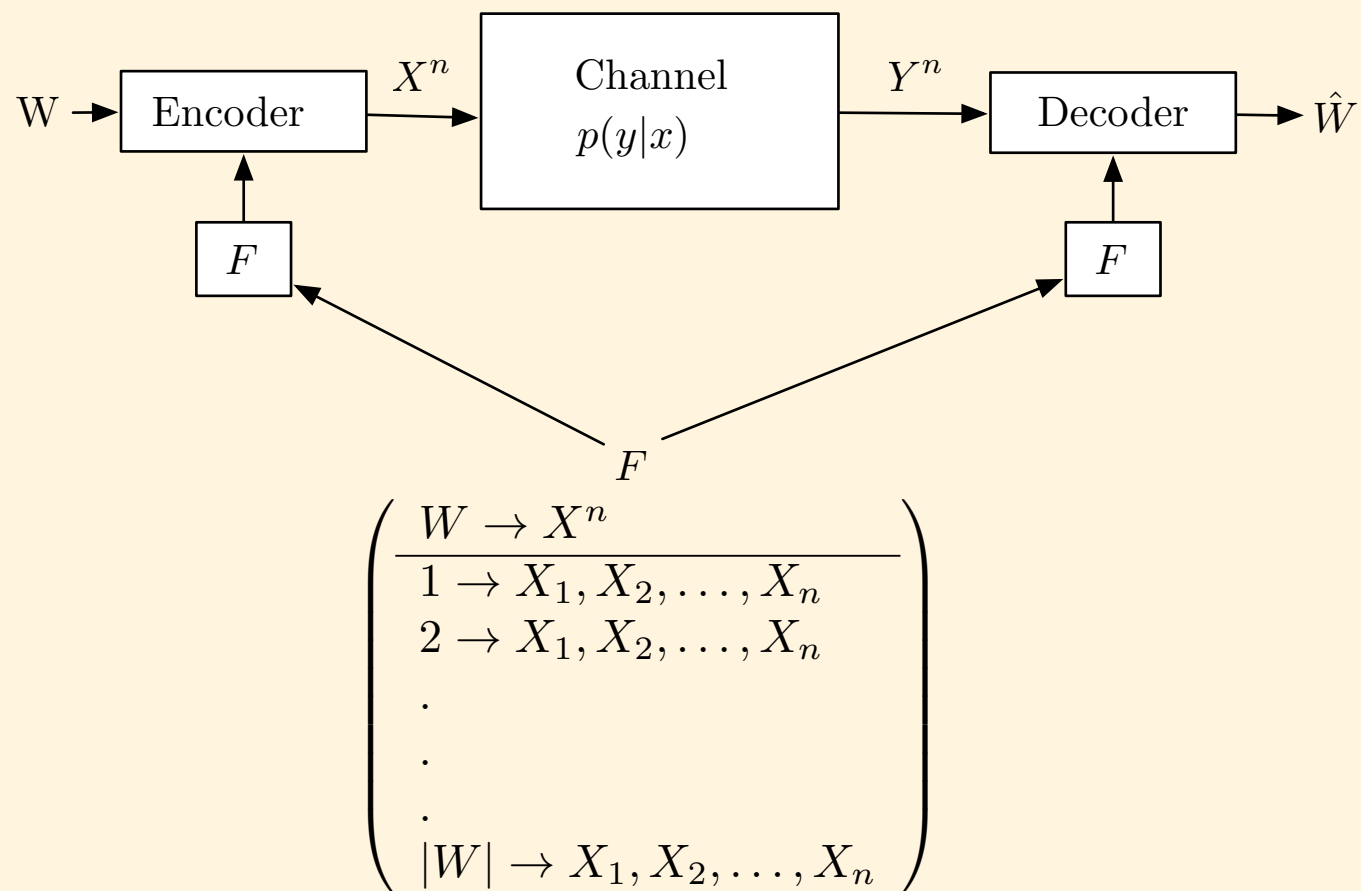


Interference Channel (IC)

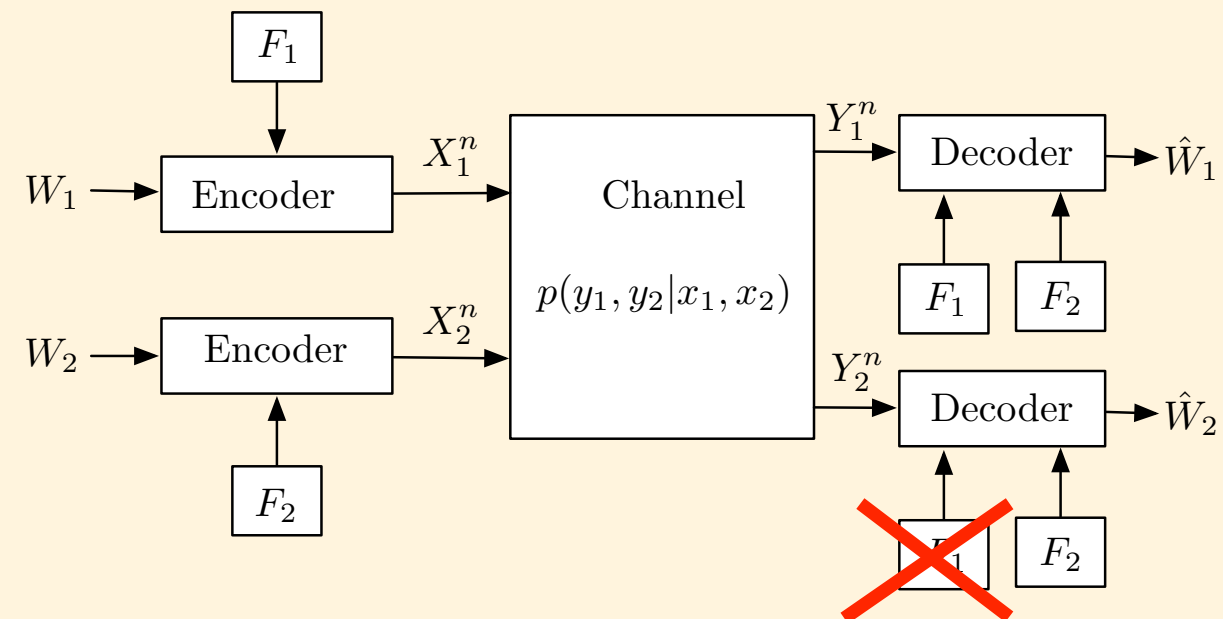


Codebook Knowledge

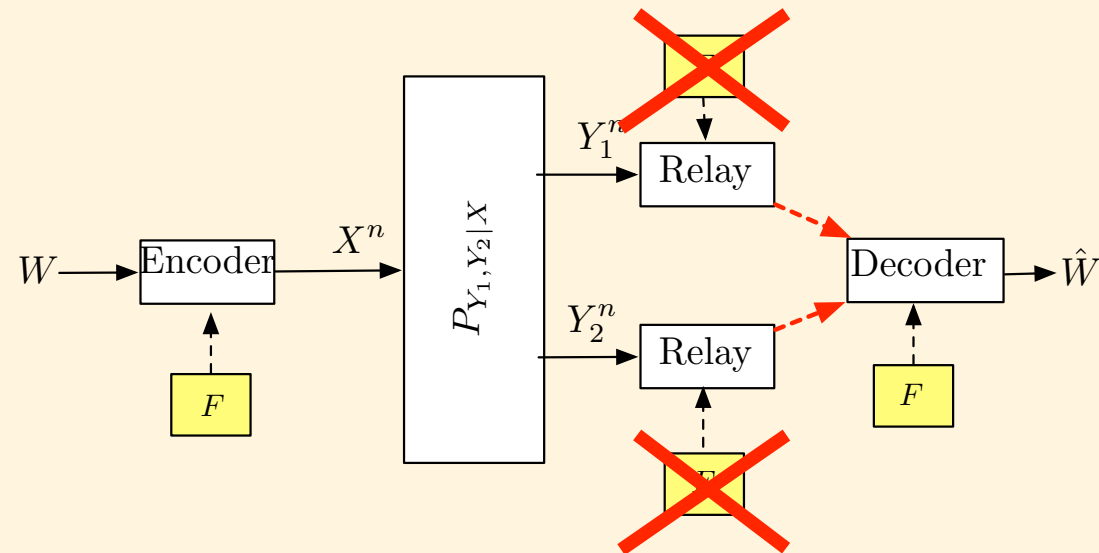
Point-to-Point Channel



Interference Channel (IC)



Past Work

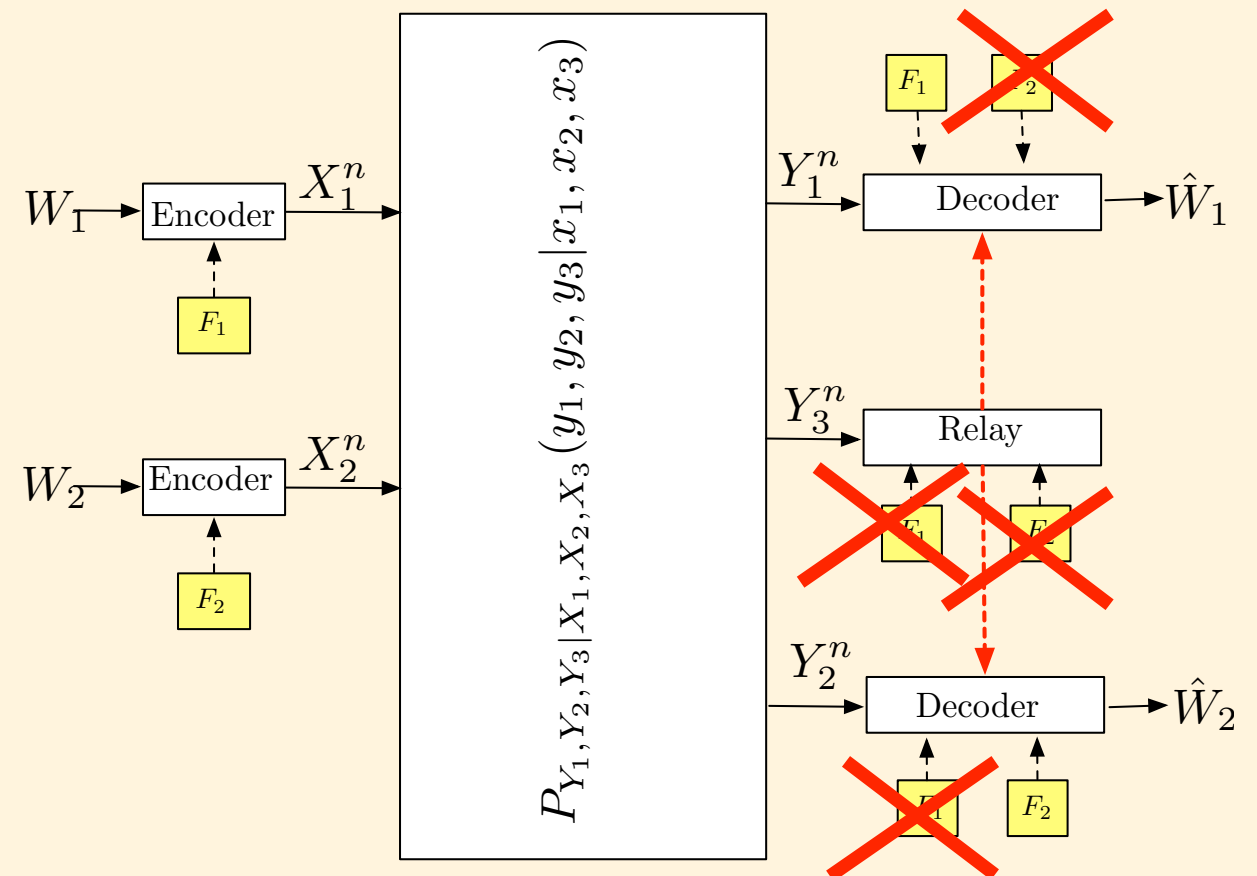


A. Sanderovich, S. Shamai, Y. Steinberg, and G. Kramer, "Communication via decentralized processing," IT July 2008.

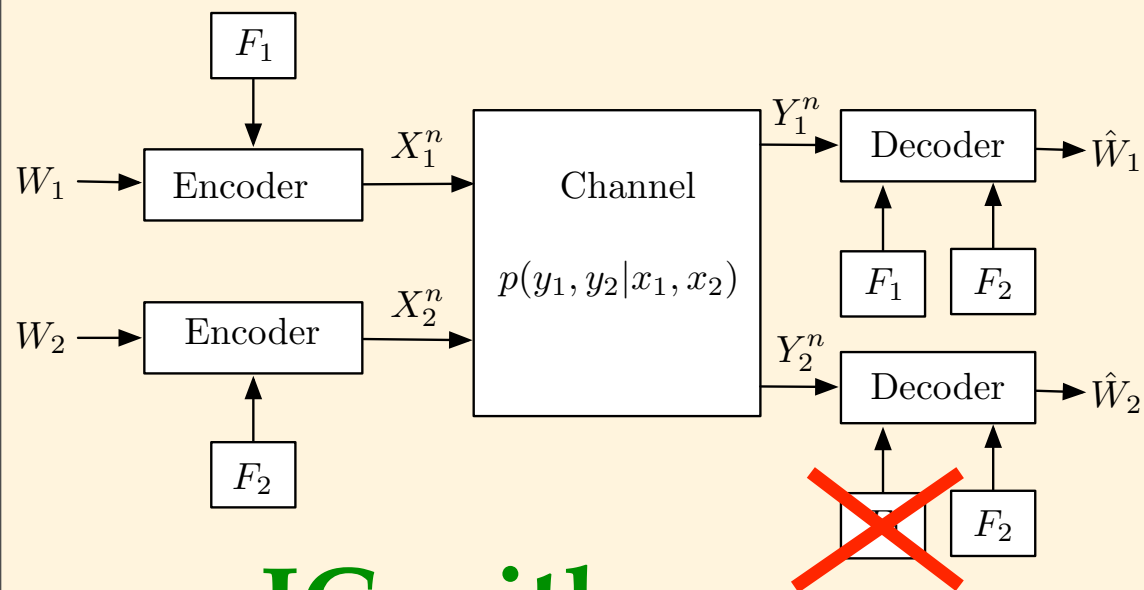
1. Upper and lower bounds, which coincide for deterministic channels
2. Gaussian noise: optimizing input unknown
3. Gaussian noise: example where BPSK outperforms Gaussian inputs

O. Simeone, E. Erkip, and S. Shamai, "On codebook information for interference relay channels with out-of-band relaying," IT May 2011.

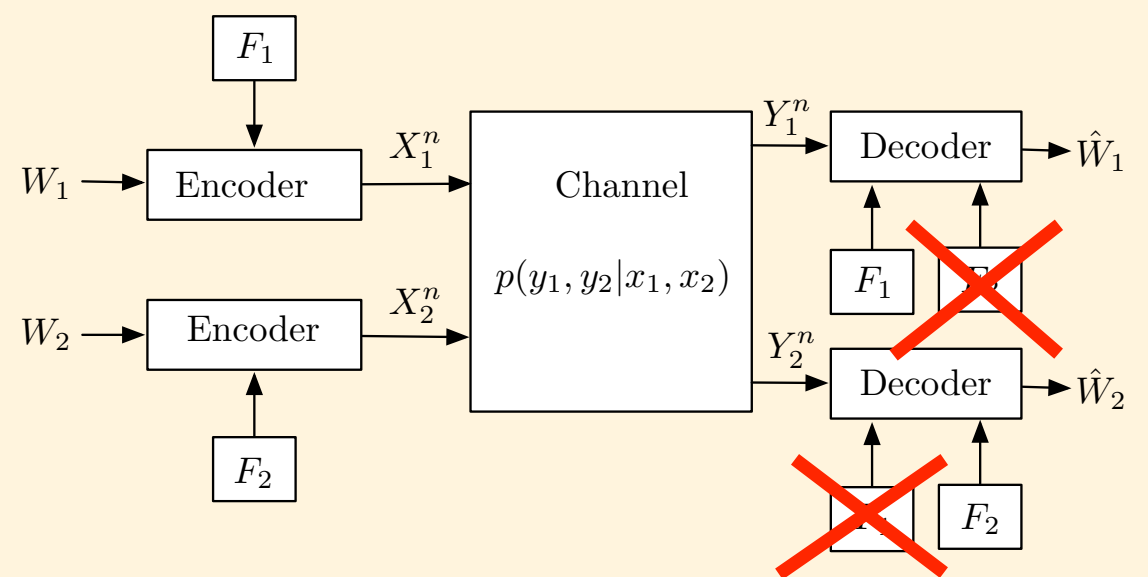
1. Primitive relay channel: capacity with compress forward
2. IC+R+Oblivious receivers: capacity with compress forward and TIAN
3. Gaussian noise: optimizing input unknown



Considered Channel Models



IC with one
oblivious Rx



IC with two
oblivious Rx

A. Dytso, N. Devroye, and D. Tuninetti, "On the capacity of interference channels with partial codebook knowledge," ISIT 2013

A. Dytso, D. Tuninetti and N. Devroye, "On Discrete Alphabets for the Two-user Gaussian Interference Channel with One Receiver Lacking Knowledge of the Interfering Codebook," ITA 2014, arXiv:1405.1117, submitted IT May 2014, revised Sept 2014

A. Dytso, D. Tuninetti and N. Devroye, "On Gaussian Interference Channels with Mixed Gaussian and Discrete Inputs," ISIT 2014

O. Simeone, E. Erkip, and S. Shamai, "On codebook information for interference relay channels with out-of-band relaying," IT May 2011

Question We Ask?

- What is the loss in performance with the loss due to lack of codebook knowledge?
- Are there inputs that do better than Gaussian? Can we provide analytical results?

Capacity Outer Bound

Theorem (Region R_0). Any achievable rate pair must satisfy

$$R_1 \leq I(Y_1; X_1 | X_2, Q)$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

$$R_1 + R_2 \leq H(Y_1 | Q) + H(Y_2 | U_2, Q)$$

$$- H(T_2 | X_2, Q) - H(T_1 | Q)$$

Holds for general DMC

Holds for ISD-IC

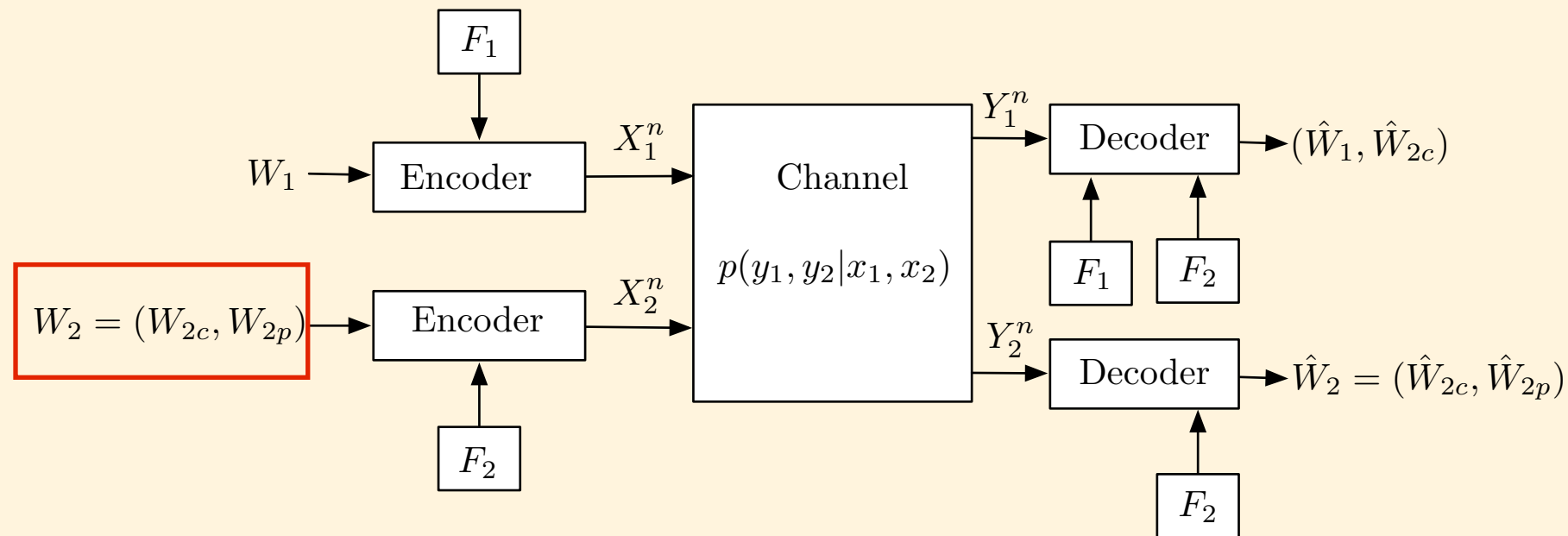
for some input distribution $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1 | Q} P_{X_2 | Q} P_{U_2 | X_2}$

where U_2 is conditionally independent copy of T_2 given X_2

$$P_{U_2, T_2 | X_2, Q}(u_2, t_2 | x_2, q) = P_{T_2 | X_2, Q}(t_2 | x_2, q) P_{T_2 | X_2, Q}(u_2 | x_2, q)$$

and where $|Q| \leq 2$

Inner Bound



Theorem (Region Ri). The following rate region is achievable

$$R_1 \leq I(X_1; Y_1 | U_2, Q)$$

$$R_2 \leq I(X_2; Y_2 | Q)$$

$$R_1 + R_2 \leq I(X_1, U_2; Y_1 | Q) + I(X_2; Y_2 | U_2, Q)$$

for all distributions in $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1|Q} P_{X_2|Q} P_{U_2|X_2}$

Special Case of HK

Gap (Con't)

Theorem. If $(R_1, R_2) \in \mathcal{R}_o$, then

$$(R_1, R_2) \in \mathcal{R}_o \implies (R_1 - \overset{\text{Gap}}{I(X_2; T_2 | U_2, Q)}, R_2) \in \mathcal{R}_i.$$

Deterministic

$$I(X_2; T_2 | U_2, Q) = 0$$

G-IC

$$I(X_2; T_2 | U_2, Q) \leq \frac{1}{2}$$

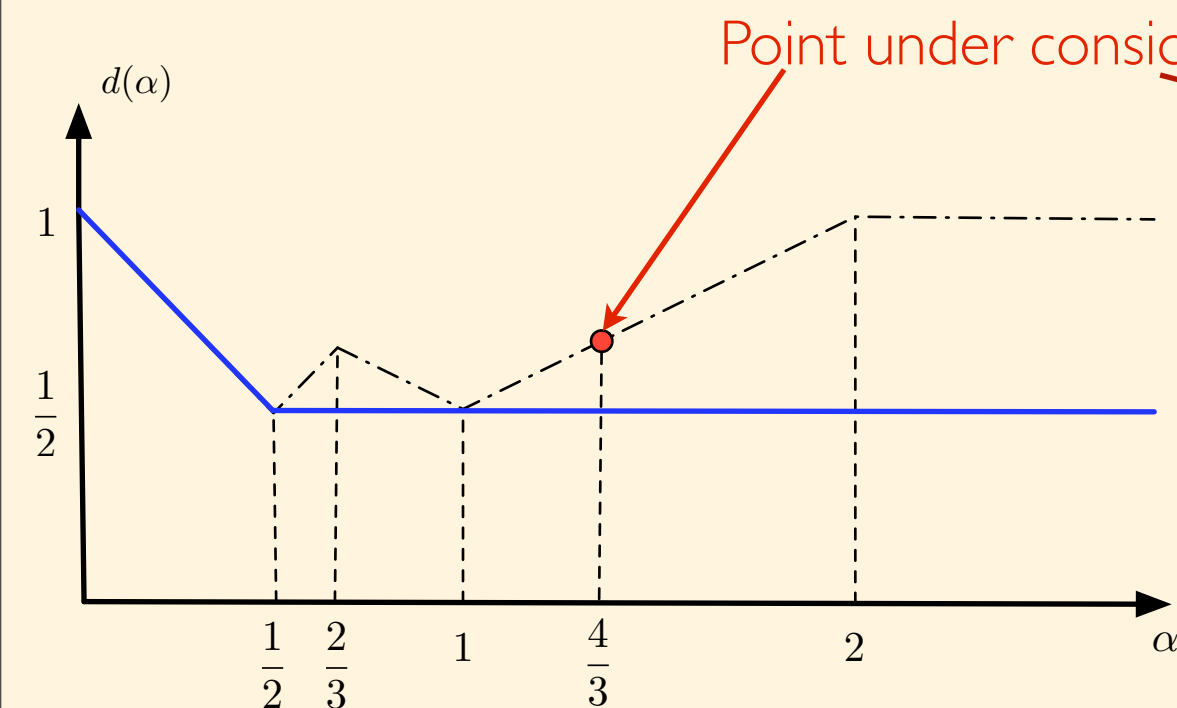
Do we know maximizing distribution?

G-IC

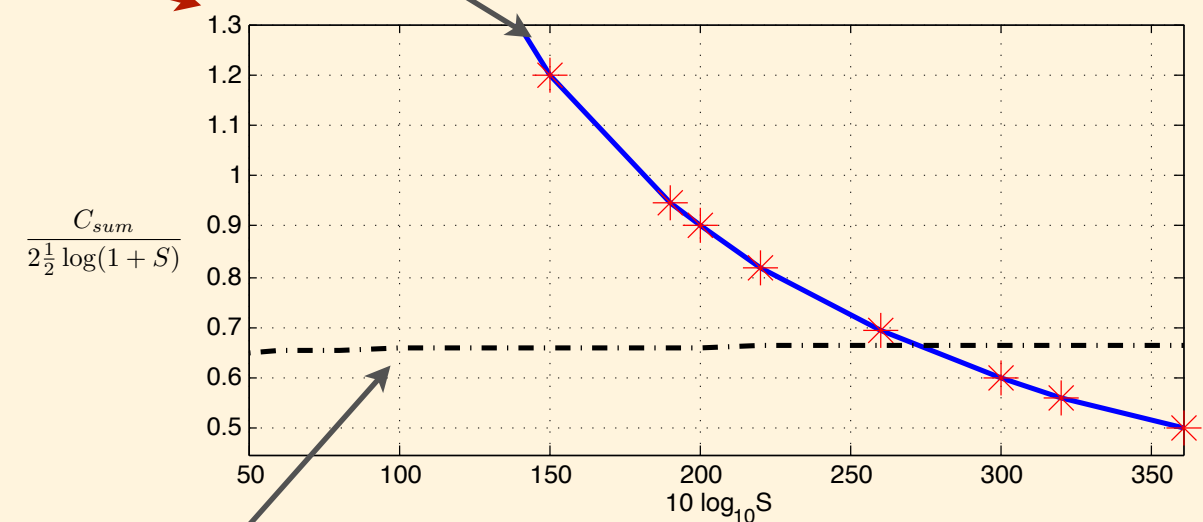
gDoF generalized Degrees of freedom

$$g(\alpha) := \lim_{S \rightarrow +\infty} \frac{C_{sum}}{2^{\frac{1}{2}} \log(1+S)}$$

$$\alpha = \frac{\log I}{\log S}$$



Both Users use Gaussian or TDMA



PAM + Gaussian

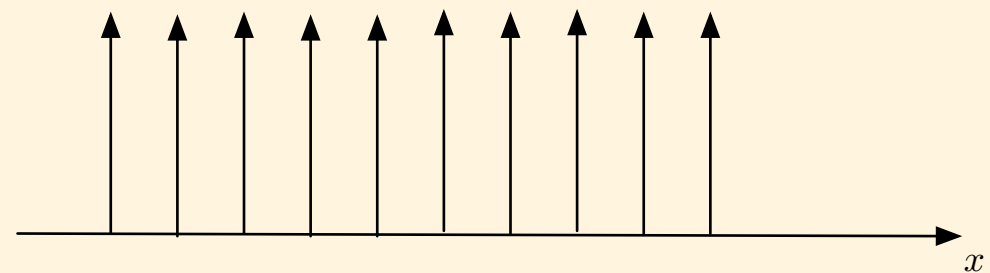
Discrete Inputs

- Discrete Input

$$X_D \sim P(X_D) = \sum_{i=1}^{|X|} p_i \delta(x_i)$$

- PAM Input

$$X_D \sim \text{PAM}(N), |X| = N, p_i = \frac{1}{N} \text{ for all } i \in [1, \dots, N]$$



- Mixed Input

$$X = \sqrt{1 - \delta} X_D + \sqrt{\delta} X_G, \delta \in [0, 1]$$

Main Tool

- A generalization of Ozarow-Wyner PAM-on-AWGN bound

$$\begin{aligned} I_d(X_D) &:= \left[H(X_D) - \text{gap}(d_{\min}(X_D)) \right]^+ \\ &\leq I(X_D; X_D + Z) \leq H(X_D), \\ \text{gap}(d) &\leq \frac{1}{2} \log \left(\frac{2\pi e}{12} \right) + \frac{1}{2} \log \left(1 + \frac{12}{d^2} \right) \end{aligned}$$

valid for any discrete input on
any additive noise channel

For PAM

$$\begin{aligned} H(X_D) &= \log(N) \\ d_{\min}(X_D) &= \sqrt{\frac{12\mathcal{E}_{X_D}}{N^2 - 1}} \end{aligned}$$

Examples (Con't)

1. Point-to-point Gaussian noise Channel

$$Y = \sqrt{\text{snr}}X + Z_G : \\ E[X^2] \leq 1, Z_G \sim \mathcal{N}(0, 1)$$

$$C = \frac{1}{2} \log(1 + \text{snr}) := I_g(\text{snr})$$

Capacity

Achievable with PAM

$$N = \lfloor \sqrt{1 + \text{snr}} \rfloor$$

$$R = \frac{1}{2} \log(1 + \text{snr}) - \text{gap}$$

$$\text{gap} = \frac{1}{2} \log \left(\frac{4\pi e}{3} \right)$$

**We have performance guarantee.
Discrete inputs are approximately
optimal.**

Example (Con't)

2. Point-to-point Gaussian noise Channel with State

$$Y = \sqrt{\text{snr}}X + hT + Z_G :$$

$$E[X^2] \leq 1, Z_G \sim \mathcal{N}(0, 1),$$

$$T \sim \text{discrete: } |T| = N \text{ and } d_{\min(T)}^2 > 0$$

$$R = I(X_G; \sqrt{\text{snr}}X_G + hT + Z_G)$$

$$\geq \frac{1}{2} \log(1 + \text{snr}) - \text{gap}$$

$$\text{gap} := \frac{1}{2} \log \left[\frac{2\pi e}{12} \left(1 + \frac{12}{d_{\min(T)}^2} \frac{|h|^2 \mathcal{E}_T}{|h|^2 \mathcal{E}_T + 1 + \text{snr}} \right) \right].$$

Interference Free Capacity

$$C = \frac{1}{2} \log(1 + \text{snr}) := I_g(\text{snr})$$

If the 'gap' is bounded then have approximately
interference free capacity

Inner Bound

$$\begin{aligned} R_1 &\leq I(Y_1; X_1 | U_2, Q) \\ R_2 &\leq I(Y_2; X_2 | Q) \\ R_1 + R_2 &\leq I(Y_1; X_1, U_2 | Q) \\ &\quad + I(Y_2; X_2 | U_2, Q) \end{aligned}$$

$X_{1D}, X_{1G}, X_{2Gc}, X_{2Gp}$ independent and distributed as

$X_{1D} \sim \text{PAM}(N), N \in \mathbb{N},$

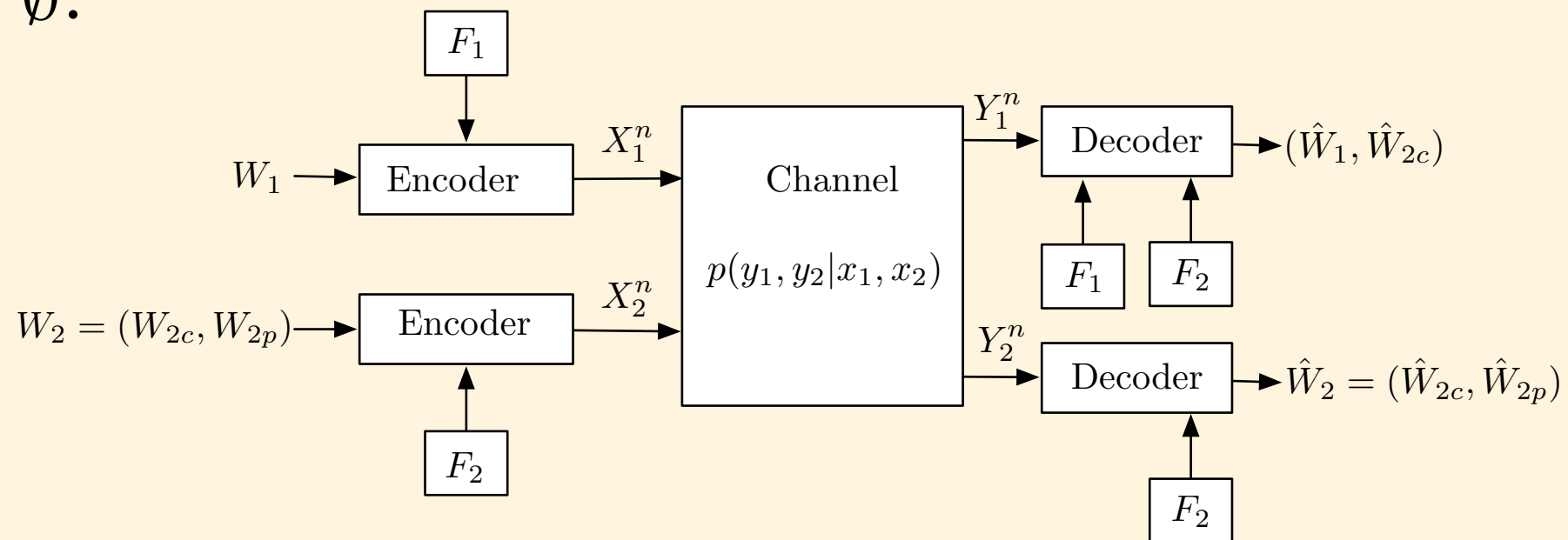
all the others are $\mathcal{N}(0, 1),$

$X_1 = \sqrt{1 - \delta_1} X_{1D} + \sqrt{\delta_1} X_{1G}, \delta_1 \in [0, 1],$

$X_2 = \sqrt{1 - \delta_2} X_{2Gc} + \sqrt{\delta_2} X_{2Gp}, \delta_2 \in [0, 1].$

$U_2 = X_{2Gc}, Q = \emptyset.$

oblivious



Inner Bound

$$R_1 \leq I(Y_1; X_1 | U_2, Q)$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

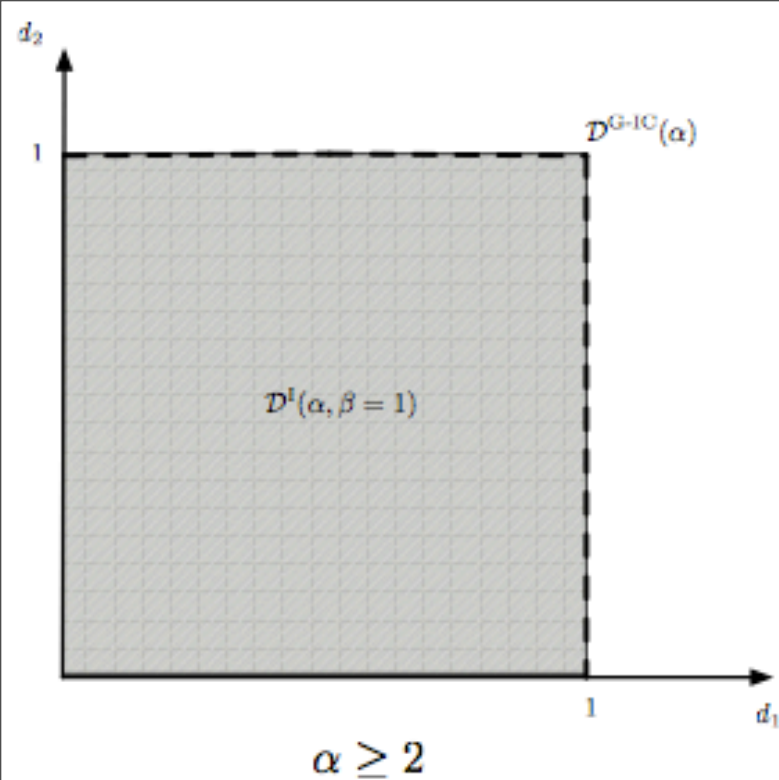
$$R_1 + R_2 \leq I(Y_1; X_1, U_2 | Q) \\ + I(Y_2; X_2 | U_2, Q)$$

$$R_1 \leq \mathsf{I}_d \left(N, \frac{|h_{11}|^2(1 - \delta_1)}{1 + |h_{11}|^2\delta_1 + |h_{12}|^2\delta_2} \right) + \mathsf{I}_g \left(\frac{|h_{11}|^2\delta_1}{1 + |h_{12}|^2\delta_2} \right),$$

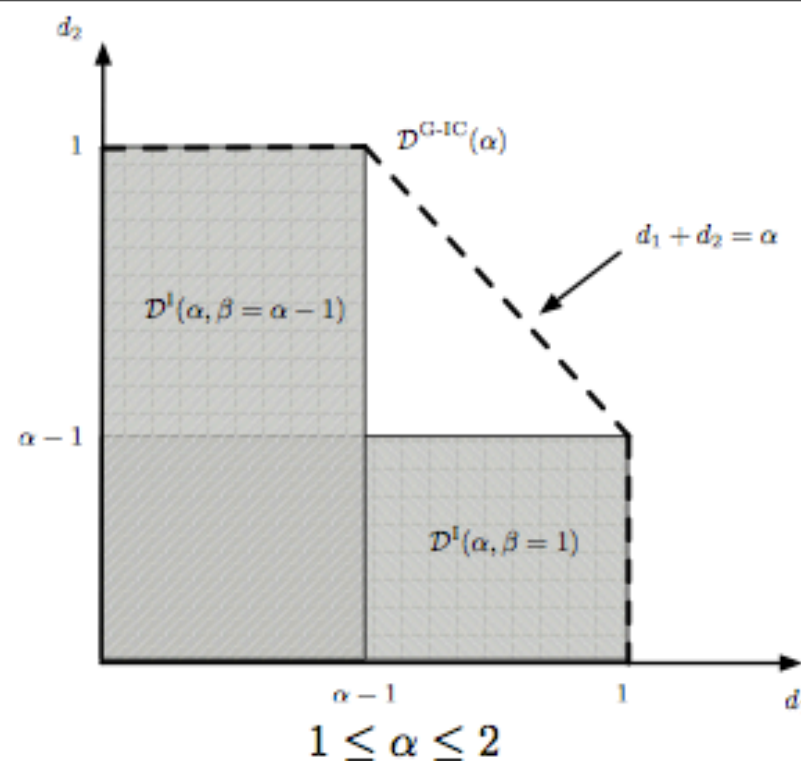
$$R_2 \leq \mathsf{I}_d \left(N, \frac{|h_{21}|^2(1 - \delta_1)}{1 + |h_{21}|^2\delta_1 + |h_{22}|^2} \right) + \mathsf{I}_g \left(\frac{|h_{22}|^2}{1 + |h_{21}|^2\delta_1} \right) \\ - \mathsf{I}_g \left(\min \left(N^2 - 1, \frac{|h_{21}|^2(1 - \delta_1)}{1 + |h_{21}|^2\delta_1} \right) \right),$$

$$R_1 + R_2 \leq \mathsf{I}_d \left(N, \frac{|h_{11}|^2(1 - \delta_1)}{1 + |h_{11}|^2\delta_1 + |h_{12}|^2} \right) + \mathsf{I}_g (|h_{11}|^2\delta_1 + |h_{12}|^2) - \mathsf{I}_g (|h_{12}|^2\delta_2) \\ + \mathsf{I}_d \left(N, \frac{|h_{21}|^2(1 - \delta_1)}{1 + |h_{21}|^2\delta_1 + |h_{22}|^2\delta_2} \right) + \mathsf{I}_g \left(\frac{|h_{22}|^2\delta_2}{1 + |h_{21}|^2\delta_1} \right) \\ - \mathsf{I}_g \left(\min \left(N^2 - 1, \frac{|h_{21}|^2(1 - \delta_1)}{1 + |h_{21}|^2\delta_1} \right) \right).$$

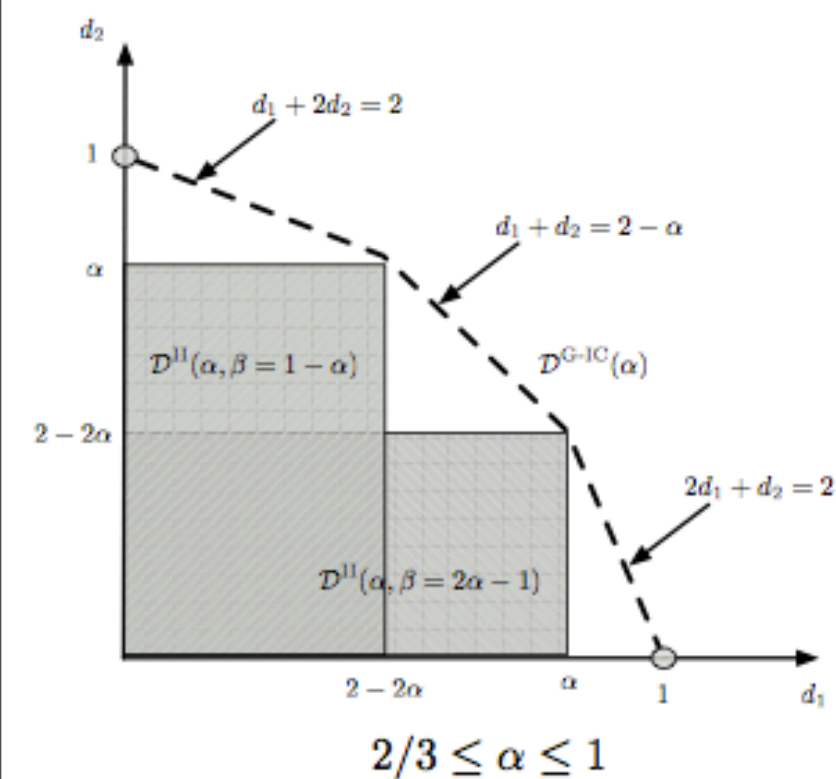
How good is this inner bound
with respect to full codebook capacity?



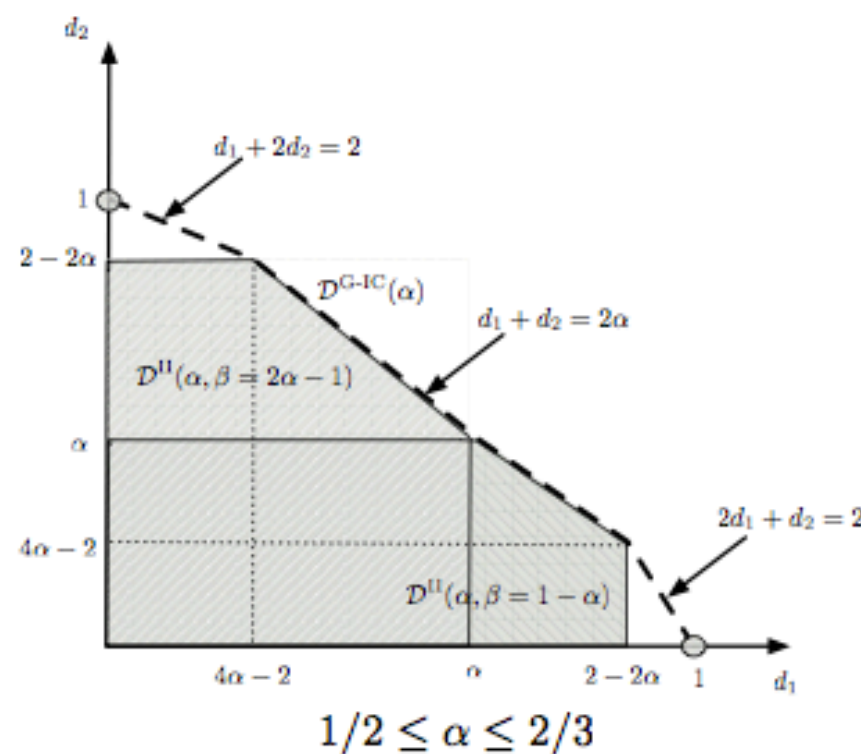
(a) Very strong interference.



(b) Strong interference.



(c) Moderately weak interference 1.



(d) Moderately weak interference 2.

$$R_1 \leq I(Y_1; X_1 | U_2, Q)$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

$$R_1 + R_2 \leq I(Y_1; X_1, U_2 | Q) + I(Y_2; X_2 | U_2, Q)$$

Power split:
either 0 (DI),
or “ETW” (DII)

$$\delta_1 = \delta_2 = \frac{1}{1 + \text{inr}}$$

$$\alpha = \frac{\log(\text{inr})}{\log(\text{snr})}$$

$$\beta = \frac{\log(N^2)}{\log(\text{snr})}$$

$$\text{gap} \leq \frac{1}{2} \log(12\pi e) \approx 3.34$$

$$N_d(x) := \lfloor \sqrt{1+x} \rfloor$$

$$\beta = 1 : N = N_d(\text{snr})$$

$$\beta = \alpha - 1 : N = N_d\left(\frac{\text{inr}}{1 + \text{snr}}\right)$$

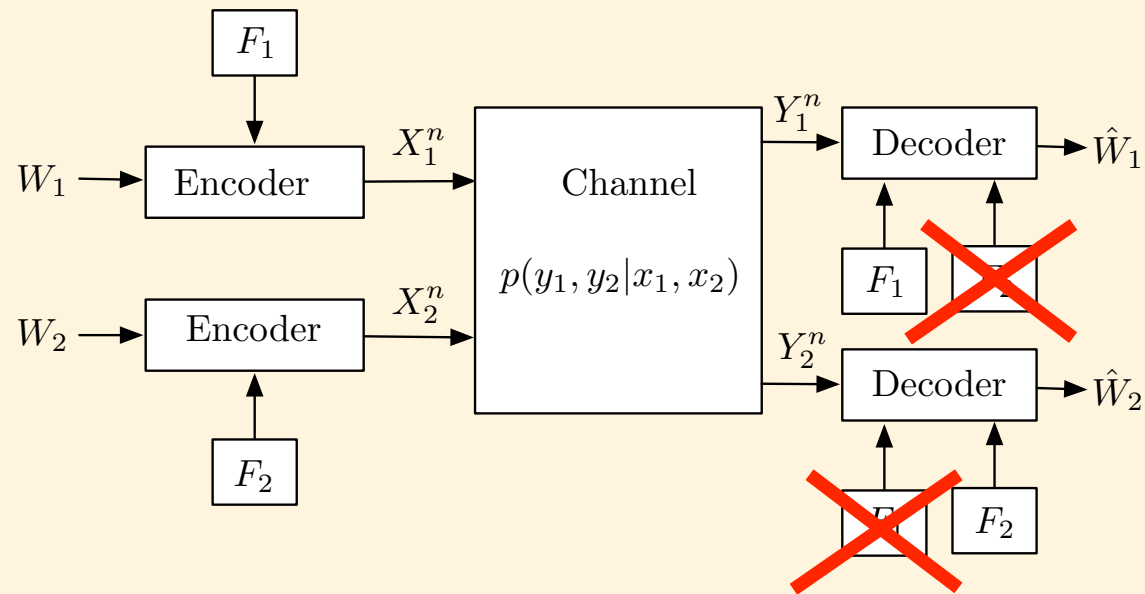
$$\beta = 2\alpha - 1 : N = N_d\left(\frac{\text{inr}^2}{1 + \text{snr} + 2\text{inr}}\right)$$

$$\beta = 1 - \alpha : N = N_d\left(\frac{\text{snr} \cdot \text{inr}}{(1 + \text{inr})^2 + \text{snr}}\right)$$

Conclusion IC-OR

- Capacity to within $1/2$ bit: do not know how to evaluate the outer bound
- Compared to outer bound with full codebook knowledge at all nodes:
PAM input at non-oblivious Tx loses at most 3.34 bits

Channel Models



Capacity

$$C = \left\{ \begin{array}{l} R_1 \leq I(X_1; Y_1 | Q) \\ R_2 \leq I(X_2; Y_2 | Q) \end{array} \right\}$$

for some distribution $p(q)p(x_1|q)p(x_2|q)$

Treating Interference as Noise (TIN) +time sharing

- Maximizing Distribution is Unknown
- TIN is very robust. Industry standard is to treat interference as Gaussian Noise.

TIN

$$C = \left\{ \begin{array}{l} R_1 \leq I(X_1; Y_1 | Q) \\ R_2 \leq I(X_2; Y_2 | Q) \end{array} \right\}$$

for some distribution $p(q)p(x_1|q)p(x_2|q)$

Look at inner bound with out time sharing

$$\mathcal{R}_{\text{in}}^{\text{TIN}} = \bigcup_{P_{X_1 X_2} = P_{X_1} P_{X_2}} \left\{ \begin{array}{l} R_1 \leq I(X_1; Y_1) \\ R_2 \leq I(X_2; Y_2) \end{array} \right\}. \quad \text{No-synchronization required}$$

Question:

$$\mathcal{R}_{\text{in}}^{\text{TIN}} = \bigcup_{P_{X_1 X_2} = P_{X_1} P_{X_2}} \left\{ \begin{array}{l} R_1 \leq I(X_1; Y_1) \\ R_2 \leq I(X_2; Y_2) \end{array} \right\} \stackrel{?}{=} \left\{ \begin{array}{l} R_1 \leq \lg(\text{snr}), \\ R_2 \leq \lg(\text{snr}), \\ R_1 + R_2 \leq \left[\lg(\text{snr}) - \lg(\text{inr}) \right]^+ + \lg(\text{inr} + \text{snr}), \\ R_1 + R_2 \leq 2\lg\left(\text{inr} + \frac{\text{snr}}{1+\text{inr}}\right), \\ 2R_1 + R_2 \leq \lg(\text{snr} + \text{inr}) + \lg\left(\text{inr} + \frac{\text{snr}}{1+\text{inr}}\right) + \left[\lg(\text{snr}) - \lg(\text{inr}) \right]^+, \\ R_1 + 2R_2 \leq \lg(\text{snr} + \text{inr}) + \lg\left(\text{inr} + \frac{\text{snr}}{1+\text{inr}}\right) + \left[\lg(\text{snr}) - \lg(\text{inr}) \right]^+, \end{array} \right\}$$

up to a constant
additive gap

Capacity outer bound with
full codebook knowledge
with synchronization

Thank you